

AKHILESH RANJAN SINGH

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Core Competencies

DevOps & Platform Engineering

Cloud Platform Engineering

Kubernetes & Container Orchestration

Infrastructure as Code (IaC) (Terraform Enterprise)

CI/CD Automation & GitOps

Multi-Cloud Architecture (GCP, AWS & Azure)

DevSecOps & Cloud Security Governance

Site Reliability Engineering (SRE)

Observability & Performance Monitoring

MLOps, LLMOps, Generative & Agentic AI Platform Engineering

IT Skills

- Cloud Platforms: Google Cloud Platform (GCP), AWS , Azure
- Infrastructure as Code: Terraform Enterprise , Ansible
- Security & Governance: IAM, RBAC, Cloud Armor, WAF, DevSecOps, Prisma Cloud, IONIX, Stream Security, SAST/DAST, Vulnerability Management
- Programming & Scripting: Python, Bash, Go, Java
- Databases: PostgreSQL, MySQL, MongoDB
- Containers & Orchestration: Kubernetes, Docker, Helm, GKE, HPA, RBAC, Network Policies
- CI/CD & GitOps: Jenkins,

Profile Summary

- Senior DevOps, SRE, and Platform Engineer with nearly 7 years of experience designing, automating, securing, and operating highly available cloud-native platforms across GCP, AWS, and Azure, with strong expertise in Kubernetes/GKE, Terraform Enterprise, CI/CD, GitOps, cloud networking, observability, DevSecOps, and production reliability.
- Proven expertise in Platform Engineering and Site Reliability Engineering, delivering enterprise GCP landing zones, self-service infrastructure, reusable Infrastructure-as-Code frameworks, Kubernetes platforms, automated deployment pipelines, observability solutions, and cloud governance controls that improve developer productivity, security, scalability, and operational reliability.
- Hands-on experience with MLOps, Generative AI, LLMOps, and Agentic AI, leveraging Vertex AI, MLflow, Gemini, RAG, LangChain, LangGraph, FastAPI, and Kubernetes to build and operate scalable AI platforms, LLM-powered applications, and agentic workflows with enterprise security and observability practices.

Work Experience

Since Sep'24| DevOps Engineer
Capgemini, Noida

Key Result Areas:

- Designing and governing enterprise-scale GCP landing zones leveraging Shared VPC architecture, project hierarchies, remote state management, IAM governance, and policy-as-code controls to ensure secure, scalable, and compliant cloud adoption.
- Developing and standardizing reusable Terraform Enterprise modules for networking, IAM, GKE, monitoring, logging, load balancing, security, and AI infrastructure, reducing infrastructure provisioning effort by 70% and accelerating platform deployment.
- Enabling self-service infrastructure delivery through integration of Terraform Enterprise with Git-based CI/CD platforms, Harness, Jira, and ServiceNow, streamlining change management and improving developer productivity.
- Administering production-grade GKE environments, managing cluster and node pool lifecycle, upgrades, autoscaling, RBAC, networking, workload scheduling, and platform operations to ensure scalability, high availability, and service reliability.
- Building and managing CI/CD and GitOps workflows using Jenkins, GitHub Actions, Harness, ArgoCD, Docker, Kubernetes, Helm, and Terraform, enabling automated and reliable application and infrastructure deployments across multiple environments.
- Driving SRE and production reliability practices through proactive monitoring, alerting, incident response, root cause analysis (RCA), capacity planning, and operational improvements to maintain highly available business-critical services.
- Enhancing cloud security and observability by implementing Cloud Armor WAF policies, rate limiting, Adaptive Protection, OpenTelemetry instrumentation, threat-intelligence integrations, and continuous attack-surface monitoring using IONIX and Stream Security.
- Driving cloud cost optimization, achieving a 20% reduction in cloud expenditure through resource rightsizing, autoscaling, capacity planning, and cloud governance practices.
- Designing and implementing MLOps workflows on GCP using Vertex AI Pipelines, Vertex AI Model Registry, MLflow, FastAPI, Docker, and Kubernetes for model deployment, versioning, experiment tracking, and lifecycle management.
- Engineering Generative AI and LLMOps workflows using Vertex AI, Gemini, RAG,

- ➔ GitHub Actions, GitLab CI/CD, Google Cloud Build, ArgoCD, GitOps, Bitbucket, Harness, JFrog Artifactory
- ➔ **Monitoring & Observability:** ELK Stack (Elastic search, Logstash, Kibana), Prometheus, Grafana, OpenTelemetry
- ➔ **MLOps & AI Infrastructure:** Vertex AI, MLflow, Kubeflow, Model Registry, Model Serving, Model Monitoring, Experiment Tracking, FastAPI, KServe, Seldon Core, TensorFlow Serving
- ➔ **Generative & Agentic AI:** Vertex AI, Gemini API, OpenAI API, RAG, Vector Databases, Embeddings, LangChain, LangGraph, AI Agents, Tool Calling, Stateful Workflows, Human-in-the-Loop, Hugging Face, Ollama, vLLM, Llama, Mistral

Education

- ➔ Bachelor of Engineering – Computer Science, RGPV, Bhopal, 2019 with 8.55 CGPA

Certifications

- ➔ Google Professional Cloud DevOps Engineer, 2025
- ➔ Google Professional Cloud Network Engineer, 2025
- ➔ Google Associate Cloud Engineer, 2024

- embeddings, vector databases, and FastAPI, enabling context-aware LLM capabilities and secure integration with enterprise applications.
- ➔ Developing Agentic AI workflows using LangGraph and LangChain, implementing tool calling, stateful workflows, conditional routing, human-in-the-loop controls, and integration with enterprise APIs and backend services.
- ➔ Implementing observability, security, and governance for AI workloads, incorporating OpenTelemetry-based tracing, structured logging, IAM/RBAC, secrets management, API authentication, and least-privilege access across MLOps, GenAI, and Agentic AI platforms.
- ➔ Leading GitHub-to-Bitbucket migration and repository standardization initiatives, improving source-code governance, CI/CD integration, engineering workflows, and operational efficiency.

Jul'22-Sep'24| Senior Software Engineer – DevOps

Tech Mahindra, Mumbai

Key Result Areas:

- ➔ Drove enterprise-scale CI/CD platform engineering using Jenkins, GitHub Actions, Terraform, Docker, Kubernetes, Helm, and Ansible, enhancing release efficiency and deployment reliability.
- ➔ Delivered Infrastructure as Code (IaC) solutions across GCP, AWS, and Azure using Terraform and Ansible, streamlining provisioning, configuration management, backup automation, and disaster recovery processes.
- ➔ Administered highly available Kubernetes-based application platforms leveraging Docker and Helm, enabling scalable application delivery and maintaining 99.9% service availability through proactive monitoring and operational excellence.
- ➔ Strengthened platform security and resilience by implementing DevSecOps controls, enterprise backup and disaster recovery solutions, observability frameworks, secure artifact management practices, and collaborating with cloud engineering teams to optimize infrastructure performance.
- ➔ Managed enterprise backup, disaster recovery, and data protection frameworks utilizing Veeam and Kasten K10, ensuring business continuity and platform resilience across cloud-native environments.
- ➔ Integrated cloud object storage (S3, GCS, Azure Blob) with CI/CD pipelines and Kubernetes workloads for artifact storage, state file management, and application data persistence.
- ➔ Implemented monitoring and observability solutions using Prometheus, Grafana, and ELK Stack, enabling proactive infrastructure monitoring, centralized logging, alerting, and faster incident troubleshooting.

Sep'19-Jul'22| System Engineer – Cloud & DevOps

TCS, Bengaluru

Key Result Areas:

- ➔ Managed production Kubernetes and GKE environments supporting business-critical applications, ensuring high availability, platform stability, and operational excellence.
- ➔ Automated CI/CD pipelines using Jenkins, Cloud Build, Terraform, and Infrastructure as Code practices, reducing deployment effort by 40% while improving consistency and delivery speed across GCP environments.
- ➔ Facilitated GCP infrastructure provisioning, cloud migration, and platform reliability initiatives through Terraform automation, observability frameworks, IAM governance, Linux administration, performance optimization, and cross-functional collaboration with development, infrastructure, and security teams.
- ➔ Managed end-to-end monitoring and alerting operations, proactively identifying system anomalies and ensuring high service availability across critical IT infrastructure and applications.
- ➔ Participated in production incident management, troubleshooting, root cause analysis (RCA), and remediation activities, collaborating with application, infrastructure, and security teams to minimize service disruption and maintain SLA compliance.